

# MIAE 215 · Embedded C++ & Physical Computing

## Week 3 Laboratory Briefing, Kit Logistics & Master Lab Schedule

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering (MIAE)  
Instructor: Prof. Brandon W. Gordon · Language: Embedded C++ (Arduino Core) · Platform: Atmel ATmega328P

### 1. Executive Summary & Week 3 Official Standing

[!IMPORTANT] **Week 3 Laboratory Status:**

- **Laboratory attendance during Week 3 is completely optional.** There are no in-person lab experiments conducted in Week 3, and **no lab assignment is handed in.**
- **Arduino Lab #1 officially starts in Week 4.**
- The official course lab portal is maintained at:  
[https://users.encs.concordia.ca/~bwgordon/MIAE\\_215\\_labs.html](https://users.encs.concordia.ca/~bwgordon/MIAE_215_labs.html).

### 2. Master Laboratory Schedule & Milestone Roadmap

The laboratory component provides hands-on practical experience in circuit interfacing, sensor signal acquisition, and servo actuation.

| MIAE 215 LABORATORY TIMELINE & MILESTONES |                                          |                     |
|-------------------------------------------|------------------------------------------|---------------------|
| Period                                    | Laboratory Module & Technical Scope      | Deliverable Type    |
| Week 3                                    | Kit Acquisition, Driver Setup & Reading  | Independent Prep    |
| Week 4                                    | Lab 1: Software Environment & Blink Test | Optional / Practice |
| Week 6                                    | Lab 2: Digital I/O & Speed Profiling     | Optional / Practice |
| Week 8                                    | Lab 3: Analog ADC & Servo Kinematics     | Optional / Practice |
| Week 9+                                   | Arduino Mechatronics Term Project        | MANDATORY (Moodle)  |
| Final                                     | Final Exam Embedded Programming (10%)    | In-Person Exam      |

### Essential Academic Regulations

1. **Deliverable Submissions:**
  - **Labs 1, 2, and 3 are NOT submitted for grading.** They are self-study practice modules with full official solution archives posted online.
  - **Only the Arduino Term Project is submitted for a course grade** via Moodle.
2. **Attendance Policy:**
  - Scheduled campus lab sessions function as open help/drop-in hours with Teaching Assistants (TAs).
  - If you complete the labs independently on your personal computer, you do not need to attend campus labs.
  - You may drop into alternate lab sections if physical space is available.

### 3. 10% Final Exam Impact:

- **Approximately 10% of the MIAE 215 Final Examination** is composed of questions drawn directly from the Arduino laboratory circuits and C++ code.

## 3. Recommended Hardware Starter Kits & Logistics

The Arduino hardware kit works with both **Windows** and **macOS**. Compatible starter kits available on Amazon Canada:

| Starter Kit Model                           | Amazon Canada Direct Link                                           | Included Key Components                                                               |
|---------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>ELEGOO UNO Project Super Starter Kit</b> | <a href="https://www.amazon.ca/B06XXYVWVJ">Amazon.ca B06XXYVWVJ</a> | Arduino Uno R3, SG90 Servo, Photoresistors, LEDs, Resistors, Breadboard, Jumper Wires |
| <b>SunFounder Starter Kit for Arduino</b>   | <a href="https://www.amazon.ca/B08B4JY95V">Amazon.ca B08B4JY95V</a> | Uno R3, Ultrasonic Sensor, SG90 Servo, LDR Light Sensors, Buzzer, Breadboard          |
| <b>Rexqualis Complete Starter Kit</b>       | <a href="https://www.amazon.ca/B07BTB3N3J">Amazon.ca B07BTB3N3J</a> | Uno R3, Dual-axis Joystick, Stepper Motor, Servo, Resistor packs, Wires               |
| <b>Kuman Complete Starter Kit</b>           | <a href="https://www.amazon.ca/B01D8KOZF4">Amazon.ca B01D8KOZF4</a> | Uno R3, Multiple Sensors, Breadboard, Jumper cables, Power Supply Module              |

[!TIP] **Minimalist Component Checklist:** If you already have electronic components, you do not need to purchase a full boxed kit. For Labs 1–3, you need:

- 1x Arduino Uno R3 (or 100% compatible clone) with USB cable.
- 1x Half-size or full-size solderless breadboard.
- Male-to-male jumper wires (pack of 20+).
- 2x 5mm LEDs and 2x 220  $\Omega$  current-limiting resistors.
- 1x Photoresistor (LDR) and 1x 10 k $\Omega$  pull-down resistor.
- 1x SG90 9g micro-servo motor.
- 1x Additional sensor of your choice for the Term Project (e.g., HC-SR04 ultrasonic distance sensor, analog 2-axis joystick, or TMP36 temperature sensor).

## 4. Software Setup & Download Resources

### 1. Arduino IDE:

- Download the latest official IDE from [arduino.cc](https://www.arduino.cc).

### 2. Video Podcasts Audio Notice:

- All lab screencast videos must be downloaded first and played using **VLC Media Player**. Built-in web browser players or Windows Media Player may drop the audio track.

### 3. RAR Archive Handling:

- Teacher files use \*.rar archive formats. Extract using 7-Zip or WinRAR (as demonstrated in Lesson #1 of the mini-course).

### 4. Early Start Recommendation:

- Students are strongly encouraged to review the online lab modules early and begin planning their Term Project well ahead of the Week 9 rush.

